Covered Call Strategies: One Fact and Eight Myths*

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A covered call is a long position in a security and a short position in a call option on that security. Equity index covered calls are an attractive strategy to many investors because they have realized returns not much lower than those of the equity market but with much lower volatility. However, a number of myths about the strategy – from why it works to why an investor should or should not invest – have surfaced, and many of them are erroneously considered "common knowledge." The authors review the underlying risk and returns of covered call strategies and dispel eight common myths about them.

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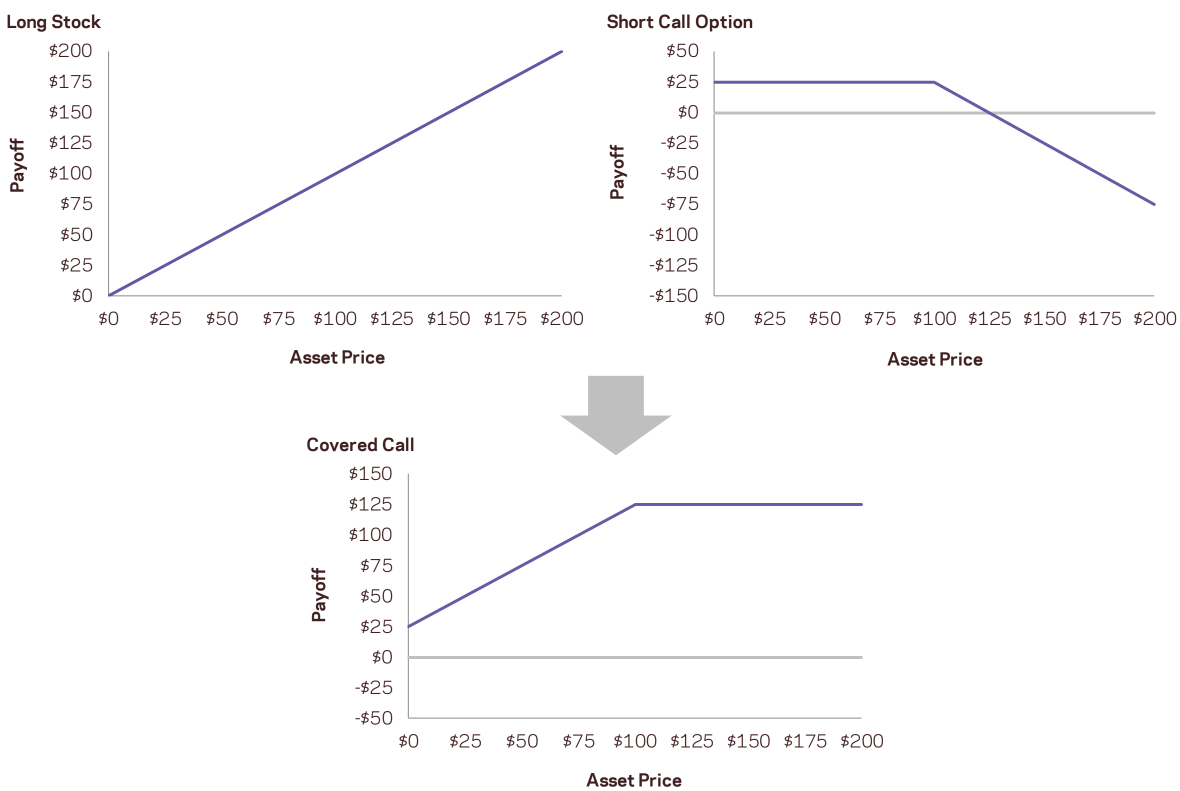
Introduction: What's in a Covered Call?

A covered call is a combined long position in a security and short position in a call option on that security. The combined position caps the investor's upside on the underlying security at the option's strike price in exchange for option premium. **Exhibit 1** constructs the covered call payoff diagram, including the option premium, at expiration when the call option is written at a $100 strike with a $25 option premium.

The covered call strategy has generated attention due to its attractive historical risk-adjusted returns. For example, the CBOE S&P 500 BuyWrite Index, the industry-standard covered call benchmark, is commonly described as providing average returns comparable to the S&P 500 Index with approximately two-thirds the volatility, supported by statistics such as those shown in **Table 1**.[1]

Although the BuyWrite Index has historically demonstrated similar total returns to the S&P 500, it does so with a reduced beta to the S&P 500 Index. However, it is important to also understand that the BuyWrite Index is more exposed to negative S&P 500 returns than positive returns. This asymmetric relationship to the S&P 500 is consistent with its payoff characteristics and results from the fact that a covered call strategy sells optionality. What this means in simple terms is that while drawdowns are somewhat mitigated by the revenue associated with call writing, the upside is capped by those same call options. **Exhibit 1** constructs the covered call payoff diagram from its underlying components.

Exhibit 1. Covered Call Payoff Diagram



[1] See Whaley (2002), Feldman and Roy (2004), and Hill, Balasubramanian, Gregory, and Tierens (2006).

Table 1. Summary Statistics

July 1, 1986—December 31, 2013

	S&P 500	CBOE BuyWrite
Annualized Excess Return	5.4%	4.4%
Annualized Volatility	18.5%	13.4%
Sharpe Ratio	0.29	0.33
Worst Drawdown	-61.7%	-43.0%
Beta to S&P 500 Index	1.00	0.67
- Upside Beta	1.00	0.63
- Downside Beta	1.00	0.78

Source: S&P 500 Total Return Index and CBOE S&P 500 BuyWrite Index. Returns are excess of cash (U.S. 3-Month LIBOR).

For obvious reasons, strategies that may offer equity-like expected returns, but with lower volatility and market exposure (beta) generate strong investor interest. As a case in point, covered call strategies are one such approach that has been gaining popularity. Growth in assets under management in covered call strategies has been over 25% per year over the past 10 years (through June 2014) with over $45 billion currently invested.[2]

Recently another strategy with similar performance objectives has piqued the interests of investors: low volatility investing. However, unlike covered call strategies, the sources of returns underlying low-volatility investing have not been subject to so many confusing and distracting myths.

Exhibit 2. One Fact and Eight Myths Summary

One Fact
Index covered calls provide long equity and short volatility exposure.
Eight Myths
1 Risk exposures can be expressed in a payoff diagram.
2 Covered calls provide downside protection.
3 Covered calls generate income.
4 Covered calls on high volatility stocks and/or shorter-dated options provide higher yield.
5 Time decay of options written works in your favor.
6 Covered calls are appropriate if you have a neutral to moderately bullish view.
7 Overwriting pays you for doing what you were going to do anyway.
8 Overwriting allows you to buy a stock at a discounted price.

Low volatility investing is based on the low risk anomaly which runs contrary to textbook finance theory: low risk stocks, as defined by their volatility, idiosyncratic volatility, or beta, do not have lower average returns than their high risk counterparts.[3] This characteristic may be used to construct a portfolio with average returns comparable to an underlying equity index, but with lower volatility. The resulting

[2] Source: Morningstar Direct and eVestment Fund Databases.

[3] Many papers have been written on this topic, including Black, Jensen, and Scholes (1972), Ang, Hodrick, Xing, and Zhang (2006 and 2009), Asness, Frazzini, and Pedersen (2012, 2013, and 2014), and Frazzini and Pedersen (2012 and 2014).

portfolio provides investors two sources of return: the equity risk premium and the low volatility anomaly.[4]

Although the low volatility anomaly portfolio is generally understood to allocate to two sources of return – the equity risk premium and low volatility excess returns – the covered call strategy is rarely described according to its two sources of return: the equity risk premium and the volatility risk premium.[5] Rather than transparently identify the covered call's compensated risk exposures, it is more common to improperly describe the strategy as a method of producing income or obtaining downside protection. This paper identifies and debunks eight such myths about covered calls. These myths are summarized in **Exhibit 2**. To begin, we suggest that investors and portfolio managers who are interested in employing a covered call strategy must first understand the following fact:

Fact: Covered calls provide long equity and short volatility exposure.

To understand this fact, consider that a short call option has negative exposure to its underlying security's return and negative exposure to its underlying security's volatility. Therefore, writing a call option to cover an existing position reduces the portfolio's exposure to the underlying security while adding short volatility exposure to the same security.

at-the-money covered call =
long equity exposure + short volatility exposure
(½ long position)   (½ short straddle)

The above equation disentangles the two distinct exposures provided by a covered call strategy: long equity and short volatility. **Exhibit 3** reconstructs the covered call payoff diagram using this long equity and short volatility decomposition.

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[4] Li, Sullivan, and Garcia-Feijoo (2014) present evidence that low volatility anomaly returns are not due to systematic risk factor exposures. Asness, Frazzini, and Pedersen (2012) demonstrate that leverage aversion may lead to higher risk-adjusted returns for less risky assets.

[5] Bakshi and Kapadia (2003) analyze delta-hedged index option returns and find evidence in favor of a volatility risk premium. Hill, Balasubramanian, Gregory, and Tierens (2006) show that covered call returns are higher because of the spread between implied and realized volatility. Bollen and Whaley (2004) shows that net buying pressure, particularly for index put options, impacts the shape of the implied volatility surface. Garleanu, Pedersen, and Poteshman (2005) show that the volatility risk premium can be explained by demand pressure for options that cannot be perfectly hedged.

Exhibit 3. Covered Call Payoff Diagram

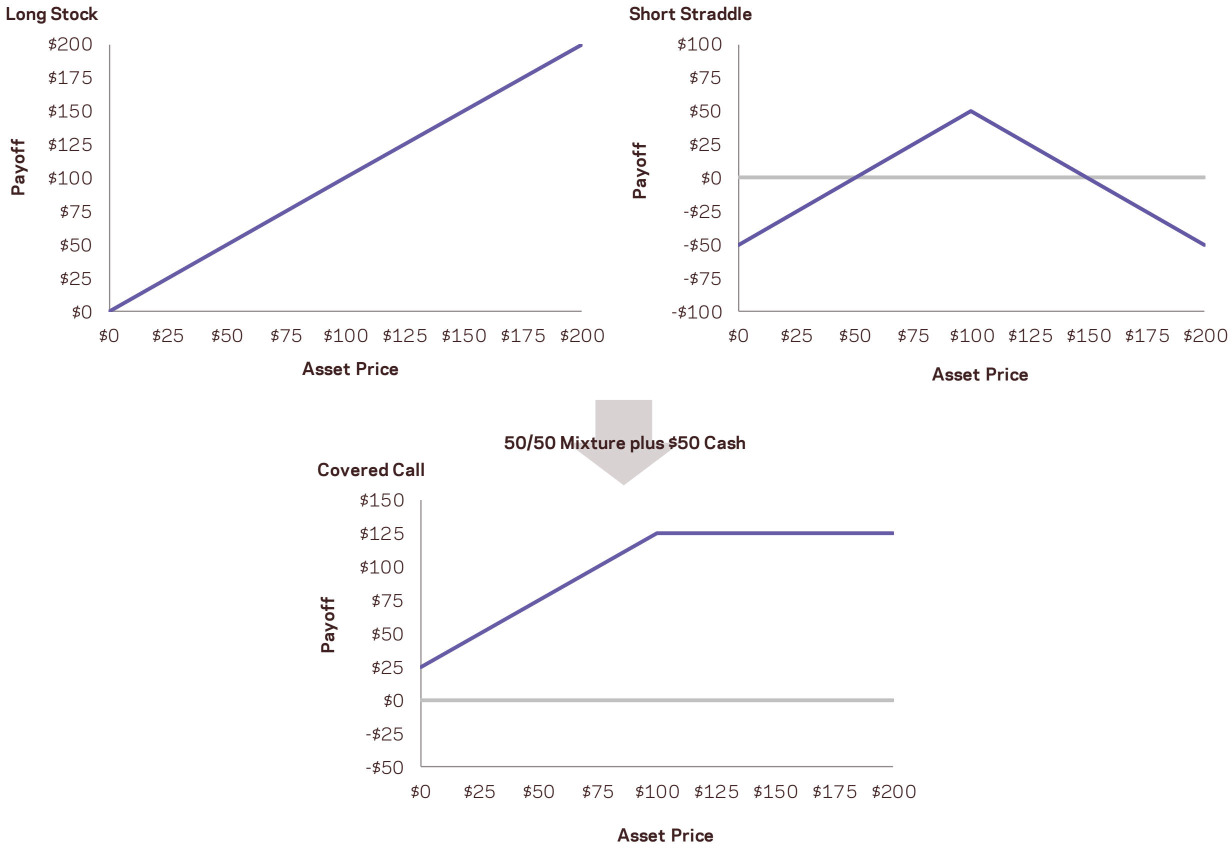


Table 2. Hypothetical Summary Statistics

April 1, 1996—December 31, 2013

	Covered Call	Long Equity	Short Straddle
Annualized Excess Return	5.0%	3.3%	1.8%
Annualized Volatility	14.1%	9.9%	6.7%
Sharpe Ratio	0.36	0.33	0.27
Skew	-0.3	0.0	-0.5
Kurtosis	20.1	8.0	21.9
Beta to S&P 500	0.64	0.50	0.14
Risk Contribution		64%	36%

Long Equity vs. Short Straddle Correlation: 0.42

The Long Equity and Short Straddle represent futures and option positions, respectively, written on the S&P 500 Index. The risk contribution is the covariance of the component with the covered call divided by the variance of the covered call.

Because options tend to be richly priced (versus the *ex ante* expected volatility of the underlying equity index) and transfer risk from option buyers to option sellers, they are considered to embed a risk premium. This risk premium is earned when selling options, which is commonly described as "selling volatility" because (as depicted in **Exhibit 3**) the short straddle position is profitable when volatility is low and unprofitable when volatility is high relative to implied volatility. For this reason, this risk premium is widely referred to as the volatility risk premium.

The short straddle component is short volatility, but it also includes additional risk due to its options' dynamic equity exposure. A short option's equity index exposure is negatively related to its equity index value in order to provide its intended payoff profile. Israelov and Nielsen (2014) show that while both short volatility exposure and dynamic equity exposure contribute to the short straddle's risk, only the volatility risk premium emanating from shorting volatility contributes to expected returns. Delta-hedging the short-straddle position substantially increases its Sharpe ratio because similar average returns are obtained at significantly lower risk.

Table 2 reports the hypothetical performance of a covered call in accordance with the above long equity and short straddle decomposition. The covered call writes one-month, at-the-money call options on option expiration dates and holds them until they expire. This strategy is similar to the CBOE BuyWrite Index except that the CBOE Index prices its sold call options using a volume weighted average price of intraday prices and our strategy writes new call positions using reported closing midpoint prices. The short straddle component is not delta-hedged.

As shown in **Table 2**, the covered call strategy is a "low beta" strategy. Writing the call option reduces the portfolio's beta from 1.00 to 0.64. The covered call's annual volatility attributed to its direct exposure to the S&P 500 is approximately 10%. Its short volatility exposure has almost 7% annual volatility, bringing the covered call's total volatility back up to 14%, which is less than the sum of the volatility parts due to the 0.42 correlation between the long equity and short straddle returns. Almost two-thirds of the strategy's total risk is allocated to the equity risk premium; the remaining one-third is allocated to the short straddle position.

The long equity component realizes returns in excess of the cash rate due to its systematic risk: the equity risk premium. The short straddle's positive performance is due to options' tendency to be richly priced: the volatility risk premium. The covered call is simply a portfolio that combines these two risk premia and the strategy's expected returns and risk are best viewed from this lens. More importantly, portfolio construction should be guided by directly targeting exposure to these two sources of return.

However, this is not the typical approach taken when constructing covered call portfolios. Instead, portfolio managers often select the strike of the written call option according to criteria such as yield target (where option premium is incorrectly classified as yield) or potential upside capture (i.e., sell options 5% out-of-the-money so that a certain amount of upside may be captured). The exposure to the actual sources of risk and return, long equity and short volatility, are a byproduct of these two criteria. We believe the direct approach of allocating the risk budget by explicitly targeting exposure to the two risk premia is more efficient.

A Stylized Example

To more clearly fix ideas on the sources of risk and return for covered call writing, we'll employ a stylized example. Consider an index with current price of $100 and a covered call written on that index with one month to maturity. For simplicity, the risk-free rate and dividend yield are assumed to be zero. We further assume the index's annualized excess return is 6%, the option-implied volatility is 18%, and realized volatility is 16%.

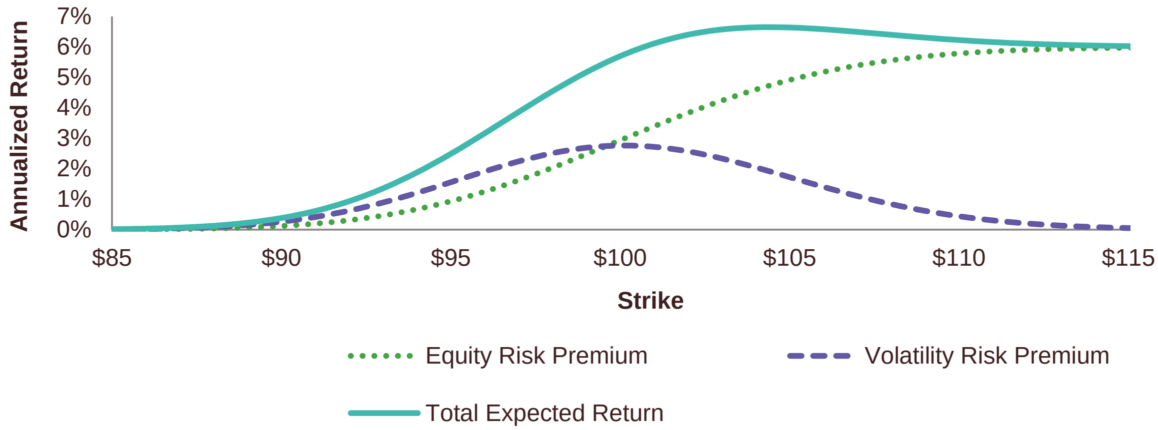
We begin with an at-the-money covered call strategy in which the call option is written at a strike equal to the current index level ($100 in this stylized example). This covered call has 0.49 exposure (delta) to the index and earns 2.94% annualized return from this exposure to the equity risk premium. The option sells for $2.07, which is $0.23 higher than it would have sold for if it were priced at expected realized volatility. In this case, approximately 11% (or $0.23/$2.07) of the collected option premium is compensation for exposure to short volatility. Annualizing the compensation earned monthly, the covered call earns 2.76% per year in volatility risk premium. The covered call's total annualized excess return is therefore 5.70% (the combined equity and volatility risk premia). This may be an attractive return considering that the at-the-money covered call has approximately 0.5 exposure to the underlying equity index.

If the option was instead priced such that its implied volatility were 16%, the same as realized volatility in our example, then even though the annual collected option premium is 22.1% of NAV, there would be zero compensation for shorting volatility (because there would be no volatility risk premium). In this case, the covered call simply earns the expected market excess return scaled by its exposure to its underlying equity (2.94% per year in our example), which is no different than what would have been earned by simply reducing the index position size by 51%. This comes as no surprise given that delta is 49%.

Selling the call option reduces exposure to the underlying equity index. The reason for selling the option, however, should be to obtain short volatility exposure in order to earn the volatility risk premium. The portfolio's equity exposure may be reduced more simply and cheaply by selling the underlying equity index. Systematically selling options only makes sense if the investor is confident that the volatility risk premium exists.

Exhibit 4 repeats the exercise for strikes between $85 and $115 and plots the return contributions due to the covered call's exposure to long equity and short volatility. This type of risk decomposition is not the norm for portfolio managers implementing covered call strategies. Instead, they tend to focus primarily on the properties of the total strategy return rather than on its components. In our stylized example, these managers would likely select a covered call written slightly out-of-the-money. For instance, the $104 strike covered call has 6.65% annualized return, with 2.05% coming from the volatility risk premium and 4.60% coming from the equity risk premium.

Exhibit 4. Stylized Representation of Underlying Risk Premia and Total Returns



We believe a better approach disaggregates expected returns to a covered call strategy by first determining the desired allocation to the two risk premia according to a set of portfolio objectives and then selecting the option strike, option leverage, and equity leverage required to achieve that objective. For instance, the at-the-money option may be preferred because it provides the highest volatility risk premium exposure per unit of leverage.

Outside of this stylized example, implied volatilities typically exhibit a "smile" (higher implied volatility for options with lower strike prices). This smile may suggest that lower strike options are more profitable to sell, and hence, a simple approach may potentially be improved upon by selling these lower strike options. Alternatively, a basket of options at different strikes may be sold in order to diversify the idiosyncratic risk embedded in each specific option.

By disentangling the short volatility and long equity exposures, the allocation to long equity and short options may be selected in order to meet risk premia allocation objectives. This optimal allocation will depend on whether the manager chooses to hedge the dynamic equity exposure described by Israelov and Nielsen (2014). Caution is in order because mean-variance optimization may not account for some adverse characteristics, specifically, losses that occur disproportionately in bad times and tails that are fatter and skewed to the downside. For this reason, selecting exposure to long equity and short volatility in order to maximize Sharpe ratio may not be the most prudent approach because the Sharpe ratio does not take into consideration the strategy's tail characteristics.

We've established that a covered call is best understood in the context of its long equity and short volatility risk exposures. The short volatility risk exposure is complex and depends non-linearly on option characteristics such as strike maturity, volatility, interest rate, and dividend yield. Potentially a result of this complexity, a number of myths on the covered call strategy have developed, leading many to invest in a good strategy for the wrong reasons. Having laid the foundation for properly assessing a covered call strategy, we can more fully address these myths and we next turn our attention to doing so.

After stating each myth, we give a one sentence plain English explanation for why it's a myth and then provide a longer discussion.

Myth 1: Risk exposure can be expressed in a payoff diagram.

Prices move before expiration and mark-to-market returns matter.

Myth 1 is true only once: at the option's expiration. The mark-to-market returns on the days prior to expiration can look very different than the payoff diagram and, as such, exposure to the underlying security may also differ significantly from the payoff diagram if the covered call position is liquidated prior to expiration.

In this regard, we can compare an option to a zero-coupon bond. The zero coupon bond's payoff diagram with respect to interest rates is a horizontal line, since we know its yield to maturity with certainty. However, we certainly do not consider a zero-coupon bond to be a riskless instrument. Instead, we realize that the bond price varies based upon a number of factors, including interest rates, inflation, and credit spreads. Consequently, in order to better understand the bond's return properties prior to its maturity, we find it useful to understand its sensitivity to interest rates (among other factors) and therefore we calculate its duration and convexity.

Where options are concerned, in a manner similar to bonds, it is helpful to first acknowledge the limitations of payoff diagrams. It is far more useful to understand the covered call's exposures to the various underlying factors that cause the option's price to change, such as the price and volatility of the underlying security. As we illustrated earlier, a covered call has long equity and short volatility exposure. Proper risk modelling and risk management requires that these two risk exposures be identified and quantified. While a payoff diagram is a useful visualization tool and helpful to understanding intuition behind options, it is of limited use in practical option risk management.

Myth 2: Covered calls provide downside protection.

Writing a call option reduces exposure to the underlying security.

A stock position can lose its full value, but a covered call can only lose the stock value less the call premium. In other words, covered calls provide some, but only very limited, support on the downside. Propagators of the downside protection myth overstate the importance of the realistic impact of the call premium on downside protection.

The root of the problem is that it is inappropriate to compare a covered call's payoff properties to that of the stock because their respective equity exposures are different. As discussed earlier, an at-the-money covered call has roughly 50 percent exposure to the stock as does a long call position. Yes, it is true that the covered call's downside is less than that of the stock, but its downside risk is significantly higher than a position in a stock at comparable leverage. Although writing the call option reduces exposure to the stock, the remaining exposure is significantly concentrated on downside risk.

To help clarify these ideas, consider a covered call position on a $100 stock with a $10 at-the-money call premium. The covered call can potentially lose $90 and the long call option can lose $10. Each position has the same 50% exposure to the stock, but the covered call's downside risk is disproportionate to its stock exposure. This is consistent with the covered call's realized upside and downside betas as reported in **Table 1**. The call seller provides insurance to the call buyer. Covered calls do not give downside protection – they provide significant downside exposure with limited upside potential.

Myth 3: Covered calls generate income.

Income is revenue minus cost.

It is true that option selling generates positive cash flow, but this incorrectly leads investors to the conclusion that covered calls generate investment income. By analogy, a zero coupon bond provides the issuer with an immediate positive cash flow in exchange for a liability. If the present value of the liability matches the sale price, then the issuance is profitless — the costs and revenues perfectly offset. It is clear that cash generated from the bond issuance is not income.

Writing a call option is similar to zero coupon bond issuance. The call option seller receives an immediate positive cash flow and a future liability. The liability obligates the option seller to sell the underlying stock at a certain price at the discretion of the option buyer, which means the stock is sold below market value if the option is exercised. Just as is the case with bond issuance, the revenue generated from selling the call option is not income (though, like income, the cash flows received from selling options are considered taxable for many investors). In order for there to be investment income or earnings, the option must be sold at a favorable price – the option's implied volatility needs to be higher than the stock's expected volatility.

Myth 4: Covered calls on high-volatility stocks and/or shorter-dated options provide higher yield.

Price is not value.

Though true that high volatility stocks and short-dated options command higher annualized premiums, insurance on riskier assets should rationally command a higher premium and selling insurance more often per year should provide higher annual premiums. However, these do not equate to higher net income or yield. For instance, if options are properly priced (e.g., according to the Black-Scholes pricing model), then selling 12 at-the-money options will generate approximately 3.5 times the cash flow of selling a single annual option, but this does not unequivocally translate into higher net profits as discussed earlier. Assuming fairly priced options, higher revenue is not necessarily a mechanism for increasing investment income.

Although it is possible that high prices are related to value, it is not necessarily the case. Myth 4 is related to Myth 3 in that it ignores the cost of the liability taken on by the option seller. The simple carry of the option may be higher (as measured by the option's time decay or proxied by the option premium), but the actual expected return is related to the option's mispricing due to a mismatch between the option's implied volatility and the stock's realized volatility as explained earlier.

In other words, expected investment profits are generated by the option's *richness*, not the option's *price*. For example, if you want to short a stock with what you consider to be a high valuation, then the goal is not to find a stock with a high price, but rather one that is overpriced relative to its fundamental value. The same principle applies to options. It is not appropriate to seek an option with a high price or other characteristics associated with high prices. Investors must instead look for options that are expensive relative to their fundamental value.

Myth 5: Time decay of options written works in your favor.

If the only thing that happened with the passage of time was the passage of time, it would be a big surprise.

It is absolutely true that an option's time value declines with the passage of time. However, this is only half (actually, less than half) the story. As time passes, prices change and volatility is realized. An option's intrinsic value increases, in expectation, as the underlying security realizes volatility. A short option position is a bet that realized volatility will be lower than implied volatility — that an option's intrinsic value will increase by less than its time value decays.

Ignoring the effect of realized volatility on an option's intrinsic value, or pretending that realized volatility will be zero, leads to the misperception that time decay is a real moneymaker. In truth, an option's time decay only works in the seller's favor if the option is initially priced expensive relative to its fundamental value. If the option is priced cheaply, then time decay works very much against the seller.

Myth 6: Covered calls are appropriate if you have a neutral to moderately bullish view.

A covered call is a bet on more than the direction of the underlying stock.

This myth is an oversimplification. In fact, when selling a call option, investors do more than merely reduce their underlying stock exposure. They are also expressing a view on the volatility of the underlying stock. Whether this is appropriate entirely depends on their view in regard to volatility and the price paid for the option as compensation to the option seller for that volatility, and has nothing to do with their view on the direction of the stock.

For instance, should investors who own a stock wish to express a negative view on its volatility (e.g., they believe that the future realized volatility will be lower than the implied volatility) while maintaining their existing view on the stock's direction, they may sell a call option and purchase additional shares of the stock to leave their stock exposure unchanged.

A neutral view on the stock may imply a belief that the security price will not move far from its current price rather than its expected return is zero. If so, then a short straddle position is a way to express that view — not a covered call — because, in this case, no active position should be taken in the security. Overwriting calls on an existing position may be appropriate if you have a neutral view and yet you are constrained from liquidating your position in the security. However, there is a trade-off because the reduced security exposure coincides with a new (risky) exposure to the security's volatility.

Myth 7: Overwriting pays you for doing what you were going to do anyway.

An option is a contractual obligation, not a plan.

This myth is typically posed as the following question: if you have a price target for selling a stock you own, why not get paid to write a call option struck at that price target?

In fact, there is an important distinction between following a plan to (hopefully) sell a stock at a certain price and being contractually obligated to do so. In the case of a long stock position, the owner plans to sell the stock at the prevailing market price when that price is equal to some pre-determined price target. However, this does not represent an obligation, only a plan, and one that can be changed in accordance with the owner's wishes. At the moment of sale, the sale price is neither favorable nor unfavorable — it is the market price at that time.

However, when selling an option struck at the same price target, the option seller is contractually obligated to sell the security at an unfavorable price (at least to the option seller) — but that is why there is an option premium. Prior to option expiration, if the underlying stock price has reached the desired price target, the long stock position may still be sold at that target price (as would have occurred had the call option not been written), but the portfolio still has the short call option position. However, the price of that short option will have changed in value since it was initially sold due to the passage of time (down), the stock price appreciation (up), and any change in the option's implied volatility (up or down). The investment risk associated with option overwriting clearly is not equivalent to getting paid to write a call option struck at some hoped for price target.

Myth 8: Overwriting allows you to buy a stock at a discounted price.

The price and value of the stock when you trade is what matters.

Although this myth is typically phrased in the context of selling naked puts, we include it in this covered call paper because selling a naked put and writing a covered call are effectively equivalent investments.

Here is how this myth is typically framed: if a stock that you would like to own is currently priced at $100 and that you think is currently expensive, you can act on that opinion by selling a naked put option at a $95 strike price and collect a premium of say $1. Then, if the price subsequently declines below the strike price, the option will likely be exercised thus requiring you to buy the stock for $95. Including the $1 premium, you effectively buy the stock at a 6% discount. If the option is not exercised you keep the premium as income. So, this type of outcome for selling naked put options may also lead you to conclude that the equivalent covered call strategy makes sense and is valuable.

This description, however is really a sleight of hand and, by the way, reflects how several other option strategies are often incorrectly presented. Why? Because this "story" gives the impression that a naked put allows the seller to obtain a favorable purchase price, better than the market price at the time the put was written, a seeming win. In fact, it is precisely the opposite. A naked put option obligates the option seller to buy the stock at the stated price at a time in which that price is unfavorable to the option seller. This is why the buyer pays an option premium in the first place.

In our example above, if the option is exercised, then when you buy the stock for $95 you won't care what the stock price was when you sold the option. What matters is the stock price on the date the option was exercised. If the stock price dropped all the way down to $80, the $95 purchase price no longer seems like a discount. Your P&L will show a mark-to-market loss of $14 ($95 - $80 - $1). The initial stock price is irrelevant and the $1 premium hardly helps.

Furthermore, if the stock price declined, as in our example, and the put option is exercised, you are more likely buying when the fundamental value is lower than on the date you sold the put option. In other words, stating today a stock's future fair price is quite simply a naïve approach to investing – prices move, but so do fundamental values. An option obligates the seller to transact at a specific price in the future, regardless of how the stock's fundamental value has changed.

Interestingly, this myth also embeds a contradiction. A naked put option (or equivalently, a covered call) provides positive exposure to the very stock that the investor chose not to purchase because he thought it was overpriced. Our above example illustrates this positive exposure; shorting a $1 naked put option that is $5 out-of-the-money loses $14 when the stock price declines $20 by the option's expiration date.

Conclusion

Call overwriting is a method of simultaneously expressing a view on a security and its volatility, and the BuyWrite Index is one of many ways to get a bundled allocation to the equity and volatility risk premia. A more informed approach requires investors to obtain their equity exposure by buying or selling the index and obtain their desired volatility exposure by buying or selling straddles.

We suggest that investors ignore the misleading storytelling about obtaining downside buffers and generating income. A covered call strategy only generates income to the extent that any other strategy generates income, by buying or selling mispriced securities or securities with an embedded risk premium. Avoid the temptation to overly focus on payoff diagrams. If you believe the index will rise and implied volatilities are rich, a covered call is a step in the right direction towards expressing that view. If you have no view on implied volatility, there is no reason to sell options.

Selling volatility is correctly considered a risky strategy. In this paper, we have attempted to demonstrate that many of the myths surrounding covered call strategies are just that: myths. In our view the myths collectively conceal the simple fact that option overwriting is a version of selling volatility. It may be a good stand-alone strategy when implied volatilities are high relative to expectations and, in particular, a good strategy when combined with earning the equity risk premium.

Related Studies

Ang, A., R. Hodrick, Y. Xing, and X. Zhang (2006), "The cross-section of volatility and expected returns", The Journal of Finance, 61 (1), 259-299.

Ang, A., R. Hodrick, Y. Xing, and X. Zhang (2009), "High idiosyncratic volatility and low returns: International and further U.S. evidence", Journal of Financial Economics, 91 (1), 1-23.

Asness, C., A. Frazzini, and L. H. Pedersen (2012), "Leverage Aversion and Risk Parity", Financial Analysts Journal, 68 (1), 47-59.

Asness, C., A. Frazzini, and L. H. Pedersen (2013), "Quality Minus Junk", Working Paper, AQR Capital Management.

Asness, C., A. Frazzini, and L. H. Pedersen (2014), "Low-Risk Investing Without Industry Bets", forthcoming Financial Analysts Journal.

Bakshi, G. and N. Kapadia (2003), "Delta-Hedged Gains and the Negative Market Volatility Risk Premium", Review of Financial Studies, 16 (2), 527-566.

Black, F., M. C. Jensen, and M. Scholes (1972), "The capital asset pricing model: some empirical tests", M. C. Jensen (Ed.), Studies in the Theory of Capital Markets, Praeger, New York, NY, 79-121.

Bollen N. P. B. and R. E. Whaley (2004), "Does Net Buying Pressure Affect the Shape of Implied Volatility Functions?", The Journal of Finance, 59 (2), 711-753.

Feldman, B. E. and D. Roy (2004), "Passive Options-Based Investment Strategies The Case of the CBOE S&P 500 Buy Write Index", Institutional Investor Journals, 2004 (1), 72-89.

Frazzini, A. and L. H. Pedersen (2012), "Embedded Leverage", Working Paper, AQR Capital Management.

Frazzini, A. and L. H. Pedersen (2014), "Betting Against Beta", Journal of Financial Economics 111 (1), 1-25.

Garleanu, N., Pedersen, L.H., and A. M. Poteshman (2009), "Demand-Based Option Pricing," Review of Financial Studies 22(10), 4259-4299.

Hill, Joanne M., V. Balasubramanian, K. Gregory, and I. Tierens (2006), "Finding Alpha via Covered Index Writing", Financial Analysts Journal, 62 (5), 29-46.

Israelov, R. and L. Nielsen (2014), "Covered Calls and Their Unintended Reversal Bet", Working Paper, AQR Capital Management.

Li, X., R. Sullivan, and L. Garcia-Feijoo (2014), "The Low-Risk Anomaly: Market Evidence on Systematic Risk vs. Mispricing", forthcoming Financial Analysts Journal.

Whaley, R. (2002), "Return and Risk of CBOE Buy Write Monthly Index", The Journal of Derivatives, 10 (2) 35-42.

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